

Cybersecurity Incident Report: Network Traffic Investigation

Prepared By

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Project Type

Network Traffic Analysis & Security Investigation

Tool Used

Wireshark

Environment

Kali Linux Virtual Machine

Part 1: Executive Summary

As part of a routine network monitoring activity, live network traffic was captured and analyzed using Wireshark to investigate communication patterns generated during web browsing sessions.

The objective of the investigation was to determine whether observed traffic represented legitimate user activity or potential indicators of compromise.

Traffic was generated through visits to the following websites:

- <https://www.kali.org/>
- <https://www.redhat.com/en>

The investigation focused on:

- DNS traffic analysis
- Protocol hierarchy review
- Conversation statistics analysis
- Traffic behavior validation

Analysis confirmed that the observed traffic was consistent with normal web browsing activity. DNS queries successfully resolved domain names, TCP sessions were established correctly, and encrypted HTTPS communications were observed throughout the capture.

No evidence of malicious communications, unauthorized external connections, or suspicious protocol activity was identified.

Part 2: Incident Description

During packet analysis, multiple network communications were observed between the local system and external web servers.

The captured traffic included:

- DNS requests and responses
- TCP session establishment
- TLS handshakes
- HTTPS communications

The purpose of the investigation was to validate that these communications were expected and did not indicate malicious behavior.

Because encrypted traffic represented the majority of network activity, additional analysis was conducted to verify communication patterns and destination behavior.

Part 3: Traffic Analysis

DNS Analysis

A display filter was applied:

```
plaintext id="z0n0cw"  
dns
```

DNS analysis showed successful hostname resolution requests generated by the browser before secure connections were established.

Observed behavior:

- Client initiated DNS queries.
- DNS server returned valid responses.
- Resolved IP addresses were subsequently used for HTTPS communications.

This behavior is consistent with legitimate browser activity.

No excessive DNS requests, suspicious domains, or unusual query patterns were observed.

Protocol Hierarchy Analysis

Protocol Hierarchy statistics were reviewed to identify the protocols responsible for the majority of captured traffic.

Observed protocols included:

- IPv4
- TCP
- DNS
- TLS/SSL
- HTTPS

Analysis showed that encrypted HTTPS communications represented a significant percentage of packet activity.

This is expected behavior when accessing modern websites.

No unusual protocols or unauthorized services were identified.

Conversation Statistics Analysis

Conversation Statistics were reviewed and sorted by total bytes transferred.

Analysis identified:

- Source and destination endpoints
- Communication sessions
- Traffic volume rankings

The highest-volume communications corresponded to browser interactions with external web servers.

Traffic flows aligned with expected client-server communication patterns.

No suspicious peer-to-peer communications or unauthorized destinations were identified.

Part 4: Analysis of Findings

Based on the packet capture and protocol analysis, the observed network activity appears to be legitimate.

Evidence supporting this conclusion includes:

- Successful DNS resolution
- Normal TCP connection establishment
- Valid TLS handshakes
- HTTPS encrypted communications
- Expected destination behavior
- Consistent browser-generated traffic patterns

No indicators suggesting malware activity, command-and-control communication, DNS tunneling, or data exfiltration were identified.

Part 5: Potential Security Risks

Although no malicious activity was detected during this investigation, similar traffic captures should be reviewed for the following indicators:

Suspicious DNS Activity

- Unknown domains
- Excessive DNS requests
- Newly registered domains
- DNS tunneling behavior

Abnormal External Connections

- Unknown IP addresses
- Unexpected geographic locations
- Unauthorized ports
- Repeated failed connections

Data Exfiltration Indicators

- Large outbound transfers
 - Persistent encrypted tunnels
 - Unusual traffic spikes
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Part 6: Recommended Immediate Containment Actions

If suspicious activity is detected during future investigations, the following containment measures are recommended.

Endpoint Isolation

Disconnect affected systems from the network to prevent:

- Malware propagation
- Lateral movement
- Data exfiltration

Block Malicious Infrastructure

Restrict:

- Malicious IP addresses
- Suspicious domains
- Unauthorized ports

using firewall and IDS/IPS controls.

Restrict Outbound Communications

Monitor and temporarily block suspicious outbound connections until further investigation is completed.

Preserve Evidence

Secure and preserve:

- Packet captures
- Firewall logs
- DNS logs
- Endpoint logs
- Security alerts

for forensic analysis.

Part 7: Recommended Investigation Procedures

Following containment, analysts should perform the following actions:

Step 1 – Review Packet Metadata

Review:

- Source IP addresses
- Destination IP addresses
- Ports
- Protocols
- Timestamps

Step 2 – Investigate DNS Activity

Analyze:

- Domain reputation
- Query frequency
- Newly observed domains

Step 3 – Validate External Connections

Review:

- Destination ownership
- ASN information
- Geographic locations
- Threat intelligence reputation

Step 4 – Correlate Security Events

Compare findings against:

- Firewall logs
- Authentication logs
- Endpoint telemetry
- SIEM alerts

Step 5 – Investigate Endpoints

Review affected systems for:

- Suspicious processes
 - Malware indicators
 - Unauthorized persistence mechanisms
-

Part 8: Supporting Tools and Logs

The following tools and logs can be used to validate network investigation findings.

Security Tools

- Wireshark
- Zeek
- NetFlow Analyzer
- Snort
- Suricata
- Splunk
- ELK Stack
- Microsoft Sentinel

Log Sources

Firewall Logs

Useful for validating:

- Connection attempts
- Allowed traffic
- Blocked traffic

DNS Logs

Useful for validating:

- Domain resolution history
- Suspicious domains
- Query frequency

Proxy Logs

Useful for validating:

- Web activity
- Download history
- External communications

Endpoint Logs

Useful for validating:

- Running processes

- User activity
- Security events

SIEM Logs

Useful for:

- Alert correlation
 - Timeline reconstruction
 - Incident validation
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Part 9: Analyst Conclusion

The investigation successfully validated network activity captured through Wireshark during controlled browser usage.

DNS requests, protocol usage, and conversation statistics were analyzed to determine whether traffic represented legitimate activity or potential security concerns.

The investigation found no evidence of malicious communications or indicators of compromise.

This exercise demonstrates practical SOC analyst skills including:

- Packet Capture
- DNS Investigation
- Protocol Analysis
- Network Monitoring
- Incident Investigation
- Security Documentation
- Traffic Interpretation

The findings indicate that the observed traffic was consistent with normal encrypted web browsing behavior and did not require incident escalation.